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Approach for Integrating a Digital Platform for Empowering Indian Farmers through Market Access, Advisory Support and Social Media Engagement

Varun Mundhada
Symbiosis Institute of Technology, Nagpur Campus
Symbiosis International (Deemed) University, Pune
varunmundhada555@gmail.com

Chinmay Vasantrao Bele
Symbiosis Institute of Technology, Nagpur Campus
Symbiosis International (Deemed) University, Pune
chinmay.bele.btech2023@sitnagpur.siu.edu.in

Jasraj Singh Ataliya
Symbiosis Institute of Technology, Nagpur Campus
Symbiosis International (Deemed) University, Pune
jasraj.ataliya.btech2023@sitnagpur.siu.edu.in

Nilesh Shelke
Symbiosis Institute of Technology, Nagpur Campus,
Symbiosis International (Deemed) University, Pune
nilesh.shelke@sitnagpur.siu.edu.in

Amit Pimpalkar
Shri Ramdeobaba College of Engineering and Management,
Ramdeobaba University, Nagpur
amit.pimpalkar@gmail.com

Rajeshwar L Balla
Symbiosis Institute of Technology, Nagpur Campus,
Symbiosis International (Deemed) University, Pune
rajubl75@gmail.com

Abstract—The **agricultural landscape in India is rapidly evolving with the integration of digital technologies aimed at improving efficiency, connectivity, and inclusivity. The proposed architecture introduces a unified, farmer-centric digital ecosystem that integrates social media-driven community engagement with services such as market linkage, advisory systems, access to government schemes, and logistics coordination. Based on post-2021 literature, government data, and usage rates, the paper will introduce the trends of digital platform usage among demographics, with peer learning, accessibility, and trust as possible parameters. It also ventures into the technologies of the future, such as AI, blockchain, and voice-based interfaces, and discusses their capabilities to enhance decision-making and transparency. The paper has identified major infrastructural and behavioral problems and presented an expandable model that suits the rural scenario. The results support a holistic, integrated module that views farmers as more than users; instead, they would be empowered contributors to a digital agricultural ecosystem.**

Keywords—Agriculture, Digital Ecosystem, Social Media, Market Linkage, Farmer Communities, AI in Farming, Blockchain, Technology Adoption, Unified Platform, FPO Integration

I. Introduction

The section provides a summary of the economic and social significance of Indian agriculture, as well as the growing influence of digital transformation on rural societies. Agriculture remains the backbone of the Indian economy, with over 42 per cent of the national workforce employed in the sector between 2021 and 2022 [1]. Despite its significance, structural issues such as unstable land fragmentation, low levels of mechanization, and climatic fluctuations result in low productivity and volatile income [2]. In India, the increased use of digital tools is aiding agricultural transformation. Farmers have now switched to mobile applications, SMS alerts, and online platforms to access real-time information about weather, prices, and government schemes [3]. Since more than 67 per cent of the village has internet access, forums such as WhatsApp and YouTube have become an essential source of information for the agricultural sector [4]. Digital inclusion, however, does not just require connection; it also necessitates other factors, such as user-friendly interfaces, vernacular language support, and digital literacy [5]. Despite the growing digital footprint, farmers face fragmented services and siloed applications for price discovery, advisory services, logistics, and scheme access, resulting in inefficiencies and reduced trust [6]. An integrated digital platform that consolidates these services is crucial for supporting farmers across the crop cycle [7]. This paper proposes a unified, farmer-first digital ecosystem that connects market access, peer learning, advisory support, and government schemes through a single interface. Drawing from post-2021 research, platform data, and regional insights, it outlines a scalable model designed to empower rural communities through inclusive and adaptable technologies [8].

II. Literature Review

This section examines recent developments in digital agriculture, with a focus on the evolution of ICT tools, the rise of social media in agricultural extension, and the usage patterns of platforms among Indian farmers. It draws on post-2021 academic and field-based studies to identify strengths and gaps in existing digital platforms [9]. Over the last two decades, agricultural extension in India has transitioned from radio- and TV-based advisories to dynamic ICT-based platforms that deliver real-time, localized support. Government-led initiatives such as mKisan, Kisan Suvidha, and eNAM have laid the foundation by offering weather alerts, price data, and scheme-related updates through SMS and mobile apps. Fig. 1 introduces digital transformation in India [10].

Social media has evolved into a decentralized tool for knowledge sharing. Platforms enable WhatsApp, YouTube, and Facebook to allow farmers to share crop visuals, regional tutorials, and firsthand experiences. Agri-startups and public institutions, including Krishi Vigyan Kendras (KVKs), have begun utilizing these platforms for digital outreach and

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community building through interactive sessions and the dissemination of schemes [11].

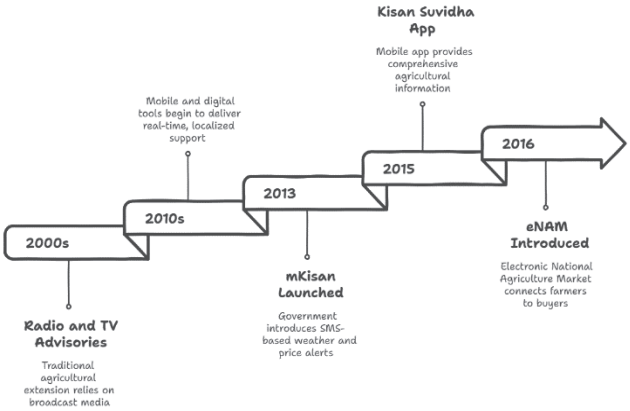

Fig. 1. Digital Transformation of Agricultural Extension in India [10]

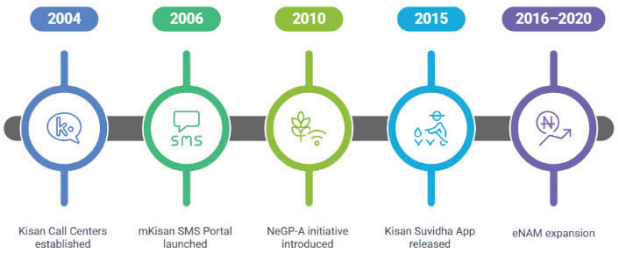

Fig. 2. ICT Innovations in Indian Agriculture (2004-2020)

TABLE. I. TIMELINE OF MAJOR ICT DEVELOPMENTS IN INDIAN AGRICULTURE

Year/ Period	ICT Initiative/Tool	Purpose	Remarks
2004	Kisan Call Centres	The toll-free number for Agri-queries	First attempt at centralized advisory
2006	mKisan SMS Portal	Weather alerts, market prices	Localized, multilingual communication
2010	NeGP-A	ICT-based agricultural governance	Created state-level Agri portals
2015	Kisan Suvidha App	Real-time information on weather and mandi	Multilingual mobile app
2016–2020	eNAM Expansion	Online mandi trading	Promoted price transparency

Table I outlines the major ICT milestones in Indian agriculture, from the introduction of Kisan Call Centres in 2004 to digital FPO initiatives post-2021 [12]. Demographics highly influence platform-specific adoption. Younger farmers (under 35) tend to prefer visual platforms like YouTube and Instagram, while older farmers rely more on WhatsApp for community-based discussions and image sharing [13]. Region-specific platforms, such as ShareChat and Moj, are gaining popularity due to their vernacular interfaces and ability to create locally relevant communities [14].

Table II summarises platform-wise usage by purpose. WhatsApp and YouTube dominate the sharing of agricultural information, while YouTube leads in news sharing, and Facebook excels in community networking.

TABLE. II. PLATFORM-WISE USAGE DISTRIBUTION BY PURPOSE AMONG FARMERS

Platform	Agricultural Info (%)	News (%)	Entertainment (%)	Peer Connection (%)
Facebook	64.5	59	48.5	76.5
WhatsApp	66.5	44.5	38	91
YouTube	76.5	69.5	86.5	23
Telegram	-	-	-	23.5
Instagram	-	-	36	34.5
Twitter	2.5	-	-	-

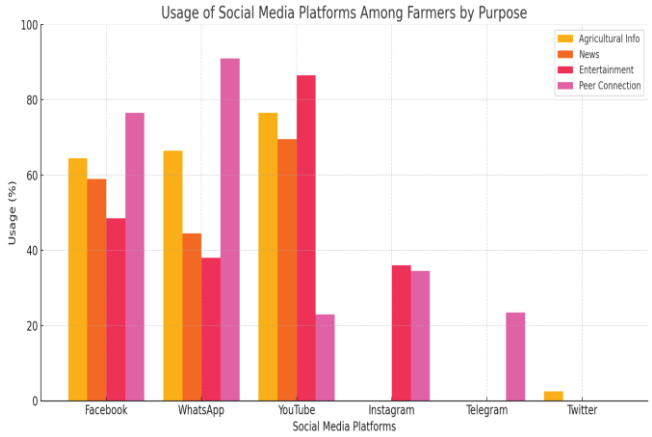

Fig. 3. Usage of Social Media Platforms Among Farmers by Purpose

As illustrated in Fig. 3, WhatsApp and YouTube dominate usage for agricultural content and informational videos, while Facebook leads in peer-to-peer networking. Telegram and Instagram serve niche functions, while Twitter remains underutilized [15].

TABLE. III. COMPARISON OF SOCIAL MEDIA PLATFORM USAGE BY AGE GROUP.

Platform	Mean Usage (Young Farmers)	Mean usage (old Farmers)	Mean Difference	Significance
Facebook	2.78	2.04	0.74	Significant
WhatsApp	2.79	2.46	0.33	Not Significant
YouTube	2.85	2.17	0.68	Significant
Telegram	1.95	1.53	0.42	Not Significant
Instagram	2.21	1.65	0.56	Significant

As shown in Table III, younger farmers exhibit significantly higher engagement across video-rich platforms, while WhatsApp remains the most universally popular platform. Social media use patterns also reflect evolving digital behaviours; younger farmers tend to use video tutorials for learning, while older users often engage in peer-led support groups. This divergence underscores the importance of platforms offering multi-format, multilingual content to ensure inclusivity and trust [16].

III. PROBLEM STATEMENT AND RESEARCH GAP

Despite increasing digital penetration in rural India, the impact of Agri-digital platforms remains limited due to fragmented service delivery, a lack of user-centric design, and minimal peer support. Most tools cater to isolated functions, market prices, crop advisories, or scheme applications, forcing farmers to juggle multiple apps, which undermines trust and usability [17].

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A. *Fragmentation of Digital Solutions*

Community-driven platforms like WhatsApp and Facebook offer informal support networks but remain largely unmoderated and unverified, increasing the risk of misinformation. The limit of structured, peer-validated systems is that they limit collective problem-solving, particularly during critical events such as pest outbreaks or policy updates [18]. Fig. 4 exhibits these barriers.

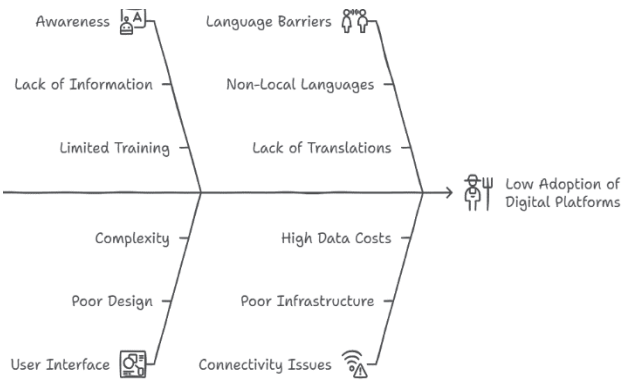

Fig. 4. Barriers to Digital Platform Adoption by Smallholders

Furthermore, awareness of digital platforms like eNAM or AgriStack remains low among smallholders. Even when services are available, poor user interfaces, language barriers, and connectivity issues hinder accessibility, particularly for low-literacy users, women, and elderly farmers [19]. There is a critical need for a holistic, integrated digital platform, as illustrated in Fig. 5, that aligns with the needs of farmers. Such a system must consolidate core services, advisory, market linkage, logistics, and schemes within a unified, multilingual, mobile-first interface. It must also promote community engagement and scalability through peer-led support, regional customization, and offline compatibility [20].

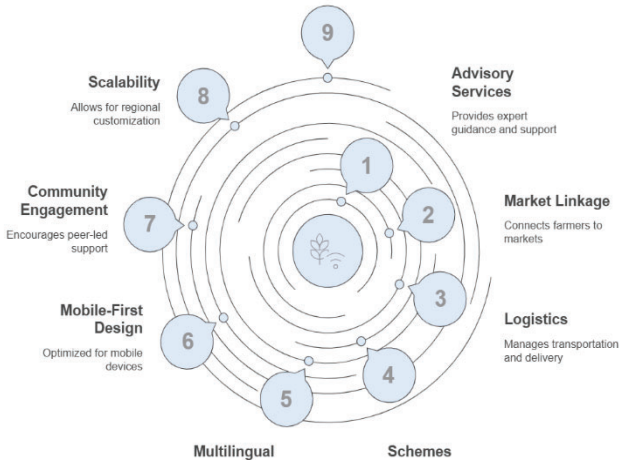

Fig. 5. Proposed Comprehensive Digital Platform for Farmers

IV. Methodology

A. *Data Collection*

This study adopts a qualitative, exploratory approach, combining primary observations with secondary research. Sources include peer-reviewed journals, government reports, and platform-specific case studies published post-2021 [21]. Surveys and platform analytics helped map user behaviours across regions, age groups, and farming contexts.

B. *Reference Dataset and Timeframes Used*

The core dataset spans 2019–2024, focusing on:

- Smartphone and internet penetration in rural India (2020–2023)
- Key digital initiatives: AgriStack, PM-KISAN, digital FPO programs
- Crop-specific data: Minimum Support Price (MSP) trends from 2010–2024 [22].

This triangulated data helped identify adoption gaps and digital preferences among Indian farmers, as shown in Fig. 6.

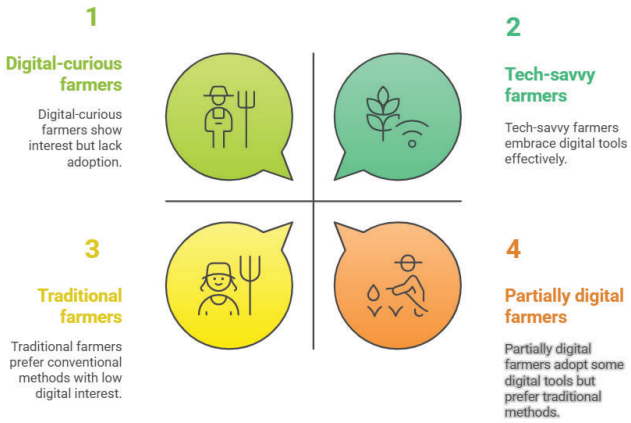

Fig. 6. Adoption Gaps and Digital Preferences in Indian Agriculture

C. *Evaluation Metrics*

Platform performance and usability, as shown in Fig. 7, were assessed based on

- Usage frequency (daily, weekly, monthly)
- Purpose (advisory, peer interaction, logistics, entertainment)
- Trust (user perception, verified content)
- Inclusivity (age, region, gender)
- Service integration and multi-functionality.

Fig. 7. Platform Performance and Usability Assessment

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D. Tools Applied

Data visualization performed using Microsoft Excel, Python (Matplotlib and Seaborn), and manual coding for thematic analysis. No proprietary APIs or external plugins were required.

The methodology employed in this study is summarises in Table V, which consolidates the data sources, analytical tools, and criteria used for evaluating digital engagement in agriculture.

TABLE. IV. SUMMARY OF METHODOLOGY

Parameter	Details
Research Type	Qualitative and secondary data-based exploratory study
Data Sources	Peer-reviewed research papers (2021–2024), government reports, and ICT platforms
Key Datasets Used	MSP data (2010–2024), social media usage reports, FPO engagement insights
Tools Applied	MS Excel (graphs), Python (Seaborn, Matplotlib), document analysis
Timeframe Considered	2019–2024 for platform data; 2010–2024 for MSP trend analysis
Evaluation Criteria	Usage frequency, purpose, reach, trust level, service integration
Geographical Focus	India (with comparative case studies from Kenya, Uganda, the EU, and the US)

V. SYSTEM DESIGN AND CONCEPTUAL FRAMEWORK

This section outlines the architectural model and major functional blocks of the proposed integrated digital platform for Indian farmers. To overcome fragmented service delivery, the proposed system adopts a modular, mobile-first architecture. It integrates core services, market linkage, advisory, logistics, financial support, and peer engagement into a unified digital platform optimized for low bandwidth and regional accessibility [23], as shown in Figs. 8-9.

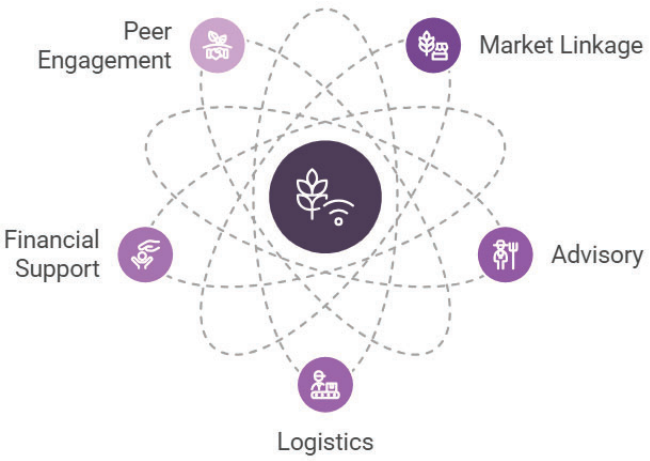

Fig. 8. Integrated Digital Platform for Indian Farmers

A. Key Components

- Community and social layers feature localised chatrooms, short videos, and interactive Q&A modules, enabling peer-to-peer learning and decentralised support. Farmers can share images, raise queries, and collaborate in real-time, building trust through community engagement [24].
- The market intelligence platform offers real-time mandi prices, buyer discovery, and demand trend visualisations. Farmers can make data-informed decisions, reducing dependence on intermediaries [25].
- Advisory services, AI-powered recommendations, and expert-generated content provide region-specific guidance on crop planning, pest management, and weather alerts. Low-literacy users supported via voice-guided chatbots and localised tutorials [26].
- Logistics coordination and integrated booking for verified transporters enable real-time scheduling and route tracking, reducing post-harvest delays and minimising spoilage [27].
- The scheme and finance support system automates access to government schemes, such as PM-KISAN, PMFBY, and eNAM, via eligibility checks, form autofill, and status tracking, thereby improving transparency and reducing dependency on intermediaries [28].

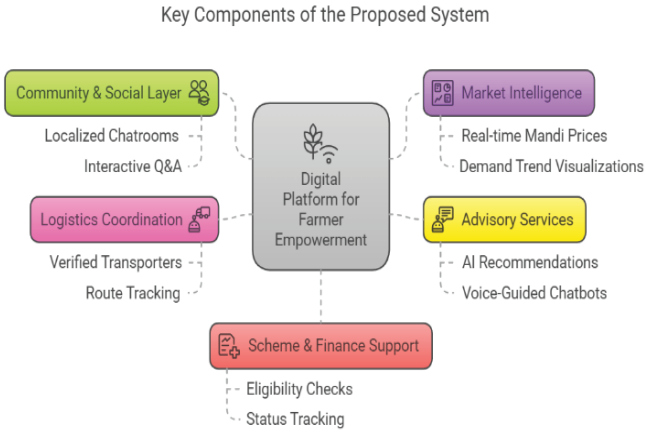

Fig. 9. Key Components of the Proposed System

E. Platform Architecture Principles

This modularity ensures adaptability across India's diverse agro-climatic and socio-economic landscape, making the platform both inclusive and future-ready.

VI. PROPOSED SOLUTION AND BENEFITS

A. MSP Trends and Economic Awareness

While the MSP for cotton has seen a general upward trend from 2010 to 2024, this has not consistently translated to income stability. Factors like rising input costs, inflation, and market fluctuations often offset MSP gains. Digitally aware farmers are better positioned to monitor pricing and align their strategies accordingly.

B. Digital Usage Patterns by Crop and Region

Analysis reveals that crop type and geographic region have a significant influence on platform adoption. Cotton and horticulture farmers show high digital engagement, particularly in states like Maharashtra and Karnataka, where mobile infrastructure and digital literacy are relatively advanced. Digital engagement does not vary across age groups, but also across farming segments. The targeted younger farmers will be given preference to video-based tutorial sites, whereas the older groups will be emphasized on peer messaging and alerts. Local applications, such as ShareChat, are also gaining more use due to their language support.

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TABLE. V. PLATFORM USAGE BY CROP TYPE

Crop Type	Common Platforms	Primary Purpose	Adoption Level
Cotton	WhatsApp, YouTube	Price trends, pest alerts, peer updates	High (Maharashtra, Gujarat)
Paddy	Facebook, WhatsApp	Seed info, weather alerts	Moderate (Eastern India)
Wheat	YouTube, Facebook	Technique sharing, government schemes	Low to Moderate
Horticulture	Instagram, WhatsApp	Visual marketing, packaging tips	High (Karnataka, Tamil Nadu)
Pulses	WhatsApp	Scheme access, regional groups	Moderate
Sugarcane	Telegram, WhatsApp	Advisory, transport coordination	Low

C. *Summary of Insights*

- Farmers in digitally advanced regions show better platform retention and multi-purpose use.
- Peer-to-peer trust and regional content drive engagement more than official portals.
- Lack of verified content and interface complexity still deters usage among smallholders.

These observations validate the need for a unified, intuitive, and localised platform tailored to the diverse realities of agriculture.

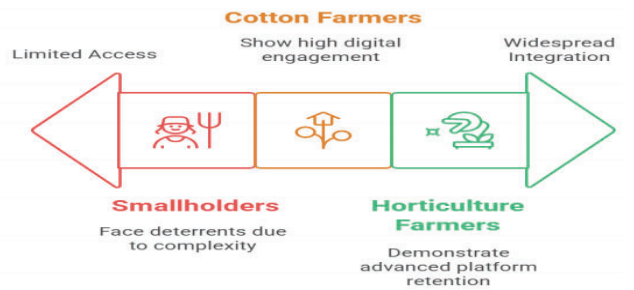

Fig. 10. Digital platform adoption among farmers based on regional digital advancement, ranging from limited access to widespread integration

VII. FUTURE SCOPE AND INNOVATION

Technology, data, and decentralization are driving a shift in India's agricultural ecosystem. In the coming decade, the digital Agri-stack is going to be even smarter, more democratic, and more farmer-driven. A combination of frontier technologies, including blockchain, voice-based artificial intelligence, and spatial analytics, provides scalable solutions that take into account the realities of smallholders. Blockchain will enable transparent supply chains and facilitate the use of smart contracts to streamline procurement and secure subsidy payments. Its decentralized structure can minimise delays, eliminate intermediaries, and enhance the market's credibility.

In the meantime, real-time, localized information will be accessible to low-literacy users, predominantly women and elderly farmers, via chatbots and voice interfaces powered by AI. Spatial mapping and drone analytics will also aid in selecting crops, monitoring soil health, and preparing irrigation. The tools can also support climate-resilient farming, where farmers can respond early to periods of extreme weather. The future platform needs to be inclusive, incorporating multilingual UI/UX, an offline-first approach, and community validators. Such improvements can foster hyper-local trust and flexibility, ensuring that none of the farmers are left behind as digital adoption gains momentum. The future scope and innovation in the field are summarized in Fig. 11.

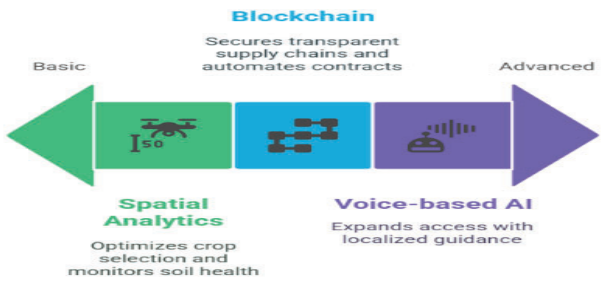

Fig. 11. Visualising the progression of technology adoption in Indian agriculture, from basic data collection to advanced, farmer-centric intelligent systems

TABLE. VI. PHASE ROADMAP

Phase	Timeline	Innovation/Feature	Objective
Phase 1	Year 1 (Pilot)	Regional language UI + Core modules	Ensure basic usability and access across literacy levels
Phase 2	Year 2	AI-based weather, crop, and pest predictions	Enable data-driven advisory and early warning systems
Phase 3	Year 2–3	Voice-based navigation and chatbot integration	Support low-literacy and elderly farmer adoption
Phase 4	Year 3	Blockchain-backed supply chain tracking	Build transparency and traceability for premium markets
Phase 5	Year 3–4	Dedicated FPO & Cooperative Management Tools	Enable community-led scaling and digital procurement
Phase 6	Year 4 onwards	Predictive analytics dashboards for FPOs	Empower collective strategy and data-informed decisions

Table VI presents a phased roadmap for scaling the platform over four years, highlighting the introduction of emerging technologies and stakeholder-driven expansion.

TABLE. VII. EXISTING INITIATIVES IN FARMER DIGITISATION: A DATASET AND IMPACT COMPARISON

Platform	Dataset	Core Services	Reference
ADeX	Soil, weather, and market	Data exchange, API	[29]
eNAM	Transaction logs	Market linkages, e-permit	[7, 10, 28]
BigHaat	User transaction data	Input + advisory + market	[30]
IIDS (AP/TG)	SMS/Voice logs	Text/voice advisories	[11, 30]
KisanQRS	Call logs	Automated responses	[12, 22]
SICKLE	Satellite imagery	Remote sensing insight	[31]

Table VII presents a comparative review of existing digital agriculture platforms, highlighting diverse approaches to empowering Indian farmers. Government initiatives, such as eNAM, ADeX, and IIDS, focus on providing market access, facilitating data exchange, and offering advisory services on a large scale. Private platforms such as DeHaat, AgroStar, and BigHaat integrate input supply, expert advice, and digital marketplaces. AI-based tools, such as KisanQRS, and datasets, like SICKLE, offer advanced support through automation and remote sensing. This comparison highlights the lack of an integrated solution that combines market access,

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advisory services, and social media engagement, which our proposed platform addresses.

VIII. Conclusion

India's agricultural sector remains central to food security, employment, and rural livelihoods. However, despite numerous digital interventions and government-backed schemes, small and marginal farmers continue to experience low productivity, service fragmentation, and limited access to digital infrastructure. This study examined how an integrated, farmer-centric digital ecosystem can bridge these gaps by embedding social interaction, advisory support, market access, logistics coordination, and scheme integration into a unified platform. Evidence from recent literature and platform-level data reveals a growing trust in peer-driven communication models such as WhatsApp groups and YouTube tutorials, signaling the relevance of community-first approaches to digital adoption. The proposed solution addresses this trend by introducing a mobile-first, multilingual interface that fosters collaborative learning, facilitates real-time problem-solving, and promotes localised engagement. The platform is designed with scalability and innovation at its core, featuring AI-powered personalized advisories, blockchain-based supply chain traceability, and voice-enabled interfaces. These features ensure that the system remains accessible, inclusive, and adaptable across diverse agro-climatic zones. Additionally, collaboration with FPOs and KVKs ensures the delivery of services to the last mile, promotes digital literacy, and fosters community trust.

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